

11(a)	intensity: vary filament current/p.d. across filament	B1
	hardness: vary accelerating potential difference	B1
11(b)(i)	$I = I_0 e^{-\mu x}$	C1
	$I_S = I_0 \exp(-0.92 \times 9.0)$ $= 2.5 \times 10^{-4} I_0$	A1
11(b)(ii)	$I_C = [\exp(-0.92 \times 6.0) \times \exp(-2.9 \times 3.0)] I_0$	C1
	$= 6.7 \times 10^{-7} I_0$	A1
11(c)	conclusion consistent with values in (b)(i) and (b)(ii) e.g. $I_S \gg I_C$ so good contrast	B1